

Supermarket Sales Performance & Customer Analytics

Project Report — based on the supplied analysis notebook

This report summarizes the analysis contained in the supplied notebook. It does not add external data or claims beyond the notebook's documented results.

1. Project Scope

The project performs an end-to-end retail analytics workflow: data loading and profiling, quality checks, cleaning and validation, feature engineering, KPI analysis, geographic performance, product and category analysis, customer and payment analysis, time-series analysis, customer satisfaction, correlation, outlier analysis, statistical testing, cross-segment analysis, business findings, recommendations, and dashboard-ready exports.

2. Executive KPIs

Metric	Value
Total sales	■244,411.08
Transactions	500
Total quantity sold	2,768
Average transaction value	■488.82
Average unit price	■88.66
Average rating	3.99 / 5
Unique products / categories	20 / 8
Unique branches / cities	4 / 4

3. Data Quality and Validation

The notebook reports zero missing values, zero duplicate rows, zero non-positive quantities, zero non-positive unit prices, zero ratings outside 1–5, and zero non-positive sales. It also verifies that recorded sales equal Quantity × Unit Price within the stated tolerance, with maximum and mean absolute differences of 0.00.

4. Main Analytical Findings

- Mumbai has the highest recorded city sales: ■72,469.45, representing 29.65% of total sales.
- Beverages is the highest-sales category at ■56,108.24.
- Cheese is the highest-sales product at ■27,906.30.
- Member customers account for ■143,009.30, or 58.51% of sales.
- UPI has the highest recorded payment-method sales at ■67,910.33.
- April 2026 has the highest monthly sales in the supplied records at ■52,569.77.
- The overall average customer rating is 3.99/5.

5. Statistical Tests

Test	Statistic	P-value
Welch's independent samples t-test	-0.35	0.73
One-way ANOVA	14.70	0.00
Chi-square: Customer Type × Payment	1.03	0.80

The notebook explicitly states that the Welch t-test interpretation should use the chosen significance level and should not be treated as causal evidence. The ANOVA tests equality of mean sales across categories, while the chi-square test examines independence between customer type and payment method.

6. Data Limitations

The notebook states that the dataset contains 500 transactions covering January–July 2026, with July incomplete. It lacks cost, profit, inventory, discount, promotion, and detailed customer demographic information. Conclusions are therefore descriptive findings from the supplied transaction data, not a complete measure of profitability or long-term market behavior.

7. Recommendations Documented in the Notebook

- Monitor high-sales products and categories for availability and demand continuity.
- Compare branch/category performance regularly to identify changing sales patterns.
- Monitor Member customer purchasing behavior separately from Normal customers.
- Track payment-channel usage to support convenient payment options.
- Monitor rating trends by product and category to identify customer-experience areas for review.
- Avoid interpreting sales as profit until cost and expense data are available.
- Continue collecting data across complete months and a longer period before making seasonal conclusions.
- Add cost, inventory, promotion, customer demographic and margin data for deeper analysis.

8. Conclusion

The supplied notebook transforms supermarket transaction data into a structured retail analytics solution spanning revenue KPIs, geographic performance, product and category performance, customer segmentation, payment behavior, customer satisfaction, time-series analysis, correlation, outlier detection, statistical testing, cross-segment analysis, and dashboard-ready outputs.